

Dr. Dhruv Kumar

BIO-DATA

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Personal Details

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Academic Qualifications

#	Degree	Year	Subject	University / Institution	CGPA
1	B.E. (Hons.)	2014	Computer Science	Birla Institute of Technology and Science, Pilani	9.92/10
2	Ph.D.	2022	Computer Science & Engineering	University of Minnesota, Twin Cities, USA	4.0/4.0

Work Experience

#	Position	Institution	From	To
1	Assistant Professor	BITS Pilani	May 2024	Present
2	Adjunct Faculty	IIIT Delhi	May 2024	Present
3	Assistant Professor	IIIT Delhi	Aug 2022	May 2024
4	Post-Doctoral Researcher	Microsoft Research, Bengaluru	Mar 2022	Jul 2022
5	Ph.D. Intern	Google Cloud, California, USA	May 2019	Aug 2019
6	Software Engineer / Quant Analyst	Goldman Sachs, Bengaluru	Nov 2014	Apr 2016
7	Summer Intern	CSIR-CEERI, Pilani	May 2012	Jul 2012

Professional Recognition / Awards / Prizes / Fellowships

#	Award / Recognition	Awarding Body	Year
1	Teaching Excellence Award (AY 2023–24)	IIIT Delhi	Sep 2024
2	Teaching Excellence Award (AY 2022–23)	IIIT Delhi	Sep 2023
3	3M Science and Technology Fellowship	University of Minnesota, Twin Cities	2017–2021
4	Travel Grant	USENIX ATC	2019
5	Travel Grant	ACM SIGMETRICS	2019
6	Travel Grant	USENIX OSDI	2018
7	Merit Scholarship (Top 10 among 800 students)	BITS Pilani	2010–2014
8	Outstanding Undergraduate Thesis	BITS Pilani	2014

Publications (in Year-wise Descending Order)

Author's name in **bold**. [*Best Paper Award*] denotes prize-winning papers.

Journal Articles & Transactions

- J1.** Alakh Sharma, Gaurish Trivedi, Kartikey Singh Bhandari, Yash Sinha, **Dhruv Kumar**, Pratik Narang, and Jagat Sesh Challa. 2026. Generative Evolutionary Meta-Solver (GEMS): Scalable Surrogate-Free Multi-Agent Reinforcement Learning. **Transactions on Machine Learning Research (TMLR) 2026**.
- J2.** Raj Gupta, Harshita Goyal, **Dhruv Kumar**, Apurv Mehra, Sanchit Sharma, Kashish Mittal, and Jagat Sesh Challa. 2026. Saksham AI: Advancing AI-Assisted Coding Education for Engineering Students in India Through Socratic Tutoring and Comprehensive Feedback. **ACM Journal on Computing and Sustainable Societies (JCSS) 2026**.
- J3.** Harshita Goyal, Garima Garg, Prisha Mordia, Veena Ramachandran, **Dhruv Kumar**, and Jagat Sesh Challa. 2025. The Impact of Large Language Models on K-12 Education in Rural India: A Thematic Analysis of Student Volunteers' Perspectives. **Scientific Reports (2025)**.
- J4.** Balamurugan Palanisamy, Vikas Hassija, Arpita Chatterjee, Arpita Mandal, Debanshi Chakraborty, Amit Pandey, G. Sai Sessa Chalapathi, and **Dhruv Kumar**. 2025. Transformers for Vision: A Survey on Innovative Methods for Computer Vision. **IEEE Access 13 (2025)**, 95496–95523.
- J5.** Harshal Akolekar, Piyush Jhamnani, Vikash Kumar, Vinay Tailor, Aditya Pote, Ankit Meena, Kamal Kumar, Jagat Sesh Challa, and **Dhruv Kumar**. 2025. The Role of Generative AI Tools in Shaping Mechanical Engineering Education from an Undergraduate Perspective. **Scientific Reports 15 (2025)**, Article 9214.
- J6.** Ashutosh Sharma, **Dhruv Kumar**, Srinivas Rallapalli, et al. 2023. Wetland Functional Assessment and Uncertainty Analysis Using Fuzzy Alpha-Cut-Based Modified Hydrogeomorphic Approach. **Environmental Science & Pollution Research 30 (2023)**, 72900–72915.
- J7.** Ananya Jain, Srinivas Rallapalli, and **Dhruv Kumar**. 2022. Cloud-Based Neuro-Fuzzy Hydro-Climatic Model for Water Quality Assessment under Uncertainty and Sensitivity. **Environmental Science & Pollution Research 29 (2022)**, 65259–65275.
- J8.** Poonam Goyal, Jagat Sesh Challa, **Dhruv Kumar**, Anuvind Bhat, Sundar Balasubramaniam, and Navneet Goyal. 2020. Grid-R-Tree: A Data Structure for Efficient Neighborhood and Nearest Neighbor Queries in Data Mining. **Int. J. Data Sci. Anal. 10 (2020)**, 25–47.
- J9.** **Dhruv Kumar**, Jian Li, Abhishek Chandra, and Ramesh K. Sitaraman. 2019. A TTL-Based Approach for Data Aggregation in Geo-Distributed Streaming Analytics. **Proc. ACM MACS 3, 2 (2019)**, 29:1–29:27.
- J10.** **Dhruv Kumar**, Poonam Goyal, and Navneet Goyal. 2018. An Efficient Method for Batch Updates in OPTICS Cluster Ordering. **Int. J. Data Anal. Tech. Strateg. 10 (2018)**, 57–80.

Conference Papers

- C1.** Stuti Sinha, Himanshu Kumar, Aryan Raju Mandapati, Rakshit Sakhuja, and **Dhruv Kumar**. 2026. The Language of Bargaining: Linguistic Effects in LLM Negotiations. In **AACL-IJCNLP 2026 (Main Conference)**, also accepted at **ICLR 2026 Workshop (LLM Reasoning)**, **ACL 2026 Workshop (C3NLP)**, **ICML 2026 Workshop (Culture x AI)**, and **ICML 2026 Workshop (Pluralistic Alignment)**.
- C2.** Darshil Chauhan, Adityasinh Solanki, Vansh Patel, Kanav Kapoor, Ritvik Jain, Aditya Bansal, **Dhruv Kumar**, and

- Pratik Narang. 2026. Navigating the Reality Gap: Privacy-Preserving On-Device Continual Adaptation of ASR for Clinical Telephony. In **AACL-IJCNLP 2026 (Main Conference)**.
- C3.** Dhruv Gupta, Vasudev Majhi, Jagat Sesh Challa, Murari Mandal, and **Dhruv Kumar**. 2026. Anchored Turns: A Length-Driven Gradient in Chain-of-Thought Causal Necessity Across Multi-Turn Conversations. In **AACL-IJCNLP 2026 (Main Conference)**.
- C4.** Venkat Aniruddh Thammatam, Vasudev Majhi, Aaryan Nagda, Jagat Sesh Challa, Murari Mandal, Yash Sinha, and **Dhruv Kumar**. 2026. Method Ranking Inverts Across Targets: A Matched-Grid Evaluation of Jailbreak Attacks on Open-Weight Vision-Language Models. In **AACL-IJCNLP 2026 (Main Conference)**.
- C5.** Devanshu Sahoo, Manish Prasad, Vasudev Majhi, Jahnvi Singh, Vinay Chamola, Yash Sinha, Murari Mandal, and **Dhruv Kumar**. 2026. When Reject Turns into Accept: Quantifying the Vulnerability of LLM-Based Scientific Reviewers to Indirect Prompt Injection. In **Findings of AACL-IJCNLP 2026**.
- C6.** Shiven Sharma, Gautam Choudhary, Ashish Sam, and **Dhruv Kumar**. 2026. Benchmarking Frontier Agentic Deception in LLMs vs Humans. In **Findings of AACL-IJCNLP 2026**.
- C7.** Debdeep Sanyal, Aakash Sen Sharma, **Dhruv Kumar**, Saurabh Deshpande, and Murari Mandal. 2026. Format Collapse in Policy Optimization for LLM Reasoning. In **EMNLP 2026**.
- C8.** Soham Roy, Sarthakbrata Halder, Arya Bharaty, Vaibhav Bhaskar, Yash Sinha, **Dhruv Kumar**, Srikant Panda, and Murari Mandal. 2026. “I Strongly Suspect This Website Is a Scam”: Benchmarking PII Leakage and Detection without Defense in Autonomous Web Agents. In **EMNLP 2026**.
- C9.** Debdeep Sanyal, Umakanta Maharana, Hong Ming Tan, Yash Sinha, Shirish Karande, Mohan Kankanhalli, **Dhruv Kumar**, and Murari Mandal. 2026. OrgAccess: A Benchmark for Role-Based Access Control in Organization Scale LLMs. In **EMNLP 2026**.
- C10.** Sasank Tumpati, Devansh Agarwal, Ayush Kedia, Arjun Neekhara, Murari Mandal, Krishna Garg, Yash Sinha, Suman Gupta, and **Dhruv Kumar**. 2026. FinBalance: A Multi-Document Accounting Reconciliation Benchmark. In **Findings of EMNLP 2026**.
- C11.** Aniket Upadhyay, Aakash Trivedi, Sparsh Bajoria, Shreyanshu Ranjan, **Dhruv Kumar**, and Pratik Narang. 2026. PRAXIS: A 68-Industry Benchmark for Pragmatic-Form Bias in Suggestion Mining. In **Findings of EMNLP 2026**.
- C12.** Aniket Upadhyay, Aakash Trivedi, **Dhruv Kumar**, and Pratik Narang. 2026. Certified Cost-Optimal Triage for Suggestion Detection in Reviews. In **EMNLP 2026 Industry Track**.
- C13.** Tulika Tewari, Nalin A. G. Arachchilage, Jagat Sesh Challa, and **Dhruv Kumar**. 2026. From Trust to Compromise: Outcome-Verified LLM Phishing Simulation and Real-Time Defense. In **Association for Computational Linguistics (ACL 2026)**.
- C14.** Samyak Savi, Chavi Gupta, Shreyas Gantayet, Tanay Sodha, and **Dhruv Kumar**. 2026. Evaluating and Mitigating Misgendering in English-to-Hindi Machine Translation. In **COLM 2026**, also accepted at **ICML 2026 Workshop (Culture x AI)**.
- C15.** **Dhruv Kumar**. 2026. Oracle-Agnostic Text Quality Optimization via Prompt-Only Agentic Loops. In **CAISc 2026**.
- C16.** Vikranth Udandaraao, Noel Abraham Tiju, Muthuraj Vairamuthu, Harsh Mistry, and **Dhruv Kumar**. 2026. Roamify: Designing and Evaluating an LLM-Based Google Chrome Extension for Personalised Itinerary Planning. In **The Web Conference (WWW) 2026**.
- C17.** Pankaj Iitd, **Dhruv Kumar**, Vinayak Abrol, and Vikram Goyal. 2026. BLEND: Balanced and Leaf-Enhanced Dual Fine-Tuning for Taxonomy Completion. In **The Web Conference (WWW) 2026**.
- C18.** Aakash Trivedi, Aniket Upadhyay, Pratik Narang, **Dhruv Kumar**, and Praveen Kumar. 2026. A Hybrid Supervised-LLM Pipeline for Actionable Suggestion Mining in Unstructured Customer Reviews. In **EACL 2026 Industry Track**.
- C19.** Debdeep Sanyal, Agniva Maiti, Umakanta Maharana, **Dhruv Kumar**, Ankur Mali, C. Lee Giles, and Murari Mandal. 2025. Investigating Pedagogical Teacher and Student LLM Agents: Genetic Adaptation Meets Retrieval-Augmented Generation Across Learning Styles. In **EMNLP 2025**, 13359–13400.
- C20.** Varun Bharti, Shashwat Jha, **Dhruv Kumar**, and Pankaj Jalote. 2025. Combining Reasoning Optimised LLMs and SMT Solvers for Automated Loop Invariant Synthesis. In **AIWare 2025**. **[Best Paper Award]**
- C21.** Aditya Pathak, Rachit Gandhi, Vaibhav Uttam, Arnav Ramamoorthy, Pratyush Ghosh, Aaryan Raj Jindal, Shreyash Verma, Aditya Mittal, Aashna Ased, Chirag Khatri, Yashwanth Nakka, Devansh, Jagat Sesh Challa, and **Dhruv Kumar**. 2025. Rubric Is All You Need: Improving LLM-Based Code Evaluation With Question-Specific Rubrics. In **ICER 2025**, 181–195.
- C22.** Adithya Chittem, Aishna Shrivastava, Sai Tarun Pendela, Jagat Sesh Challa, and **Dhruv Kumar**. 2025. SAC: A Framework for Measuring and Inducing Personality Traits in LLMs with Dynamic Intensity Control. In **ICAART 2025**.
- C23.** Madhav Krishan Garg, Tejash Prasad, Tanmay Singhal, Chhavi Kirtani, Murari Mandal, and **Dhruv Kumar**. 2025. ReviewEval: An Evaluation Framework for AI-Generated Reviews. In **Findings of EMNLP 2025**, 20542–20564.
- C24.** Utkarsh Arora, Anupam Garg, Aryan Gupta, Samyak Jain, Ronit Mehta, Rupin Oberoi, Prachi, Aryaman Raina, Manav Saini, Sachin Sharma, Jaskaran Singh, Sarthak Tyagi, and **Dhruv Kumar**. 2025. Analyzing LLM Usage in an Advanced Computing Class in India. In **ACE '25**, 154–163.
- C25.** Ritvik Budhiraja, Ishika Joshi, Jagat Sesh Challa, Harshal D. Akolekar, and **Dhruv Kumar**. 2024. “It’s not like

Jarvis, but it's pretty close!"—Examining ChatGPT's Usage among Undergraduate Students in CS. In **ACE 2024**, 124–133.

- C26.** Shubhang Shekhar Dvivedi, Vyshnav Vijay, Sai Leela Rahul Pujari, Shoumik Lodh, and **Dhruv Kumar**. 2024. A Comparative Analysis of Large Language Models for Code Documentation Generation. In **AIware 2024**.
- C27.** Harshal Dev, Raj Gupta, and **Dhruv Kumar**. 2024. From Cash to Cashless: UPI's Impact on Spending Behavior among Indian Users. In **CHI EA 2024**, 168:1–168:10.
- C28.** Shikhar Sharma, Manas Mhasakar, Apurv Mehra, Utkarsh Venaik, Ujjwal Singhal, **Dhruv Kumar**, and Kashish Mittal. 2024. Comuniqua: Exploring Large Language Models For Improving English Speaking Skills. In **COMPASS 2024**, 256–267.
- C29.** Ishika Joshi, Ritvik Budhiraja, Harshal Dev, Jahnvi Kadia, Mohammad Osama Ataullah, Sayan Mitra, Harshal D. Akolekar, and **Dhruv Kumar**. 2024. ChatGPT in the Classroom: An Analysis of Its Strengths and Weaknesses for Solving Undergraduate CS Questions. In **SIGCSE 2024**, 625–631.
- C30.** **Dhruv Kumar**, Sohaib Ahmad, Abhishek Chandra, and Ramesh K. Sitaraman. 2023. AggFirstJoin: Optimizing Geo-Distributed Joins using Aggregation-Based Transformations. In **CCGrid 2023**, 414–425. **[Best Paper Award]**
- C31.** Vikram Goyal, Naveen Kumar, Sourav S. Bhowmick, Pawan Goyal, Navneet Goyal, and **Dhruv Kumar** (Eds.). 2023. Big Data and Artificial Intelligence—11th International Conference, **BDA 2023**, Proceedings. Lecture Notes in Computer Science 14418. Springer.
- C32.** Joel Wolfrath, Nikhil Sreekumar, **Dhruv Kumar**, Yuanli Wang, and Abhishek Chandra. 2022. HACCS: Heterogeneity-Aware Clustered Client Selection for Accelerated Federated Learning. In **IPDPS 2022**, 985–995.
- C33.** Rallapalli Srinivas, Matt Drewitz, Joe Magner, Ajit Pratap Singh, **Dhruv Kumar**, and Yashwant Bhaskar Katpatal. 2022. Simulating Landscape Hydrologic Connectivity in a Precise Manner Using Hydro-Conditioning. In **Advances in Computational Modeling and Simulation**, Lecture Notes in Mechanical Engineering. Springer.
- C34.** **Dhruv Kumar**, Sohaib Ahmad, Abhishek Chandra, and Ramesh K. Sitaraman. 2021. AggNet: Cost-Aware Aggregation Networks for Geo-Distributed Streaming Analytics. In **SEC 2021**, 297–311.
- C35.** Sonal Kumari, Poonam Goyal, Ankit Sood, **Dhruv Kumar**, Sundar Balasubramaniam, and Navneet Goyal. 2017. Exact, Fast and Scalable Parallel DBSCAN for Commodity Platforms. In **ICDCN 2017**, Article 14.
- C36.** Poonam Goyal, Sonal Kumari, Sumit Sharma, **Dhruv Kumar**, Vivek Kishore, Sundar Balasubramaniam, and Navneet Goyal. 2016. A Fast, Scalable SLINK Algorithm for Commodity Cluster Computing Exploiting Spatial Locality. In **HPCC 2016**, 268–275.
- C37.** Poonam Goyal, Sonal Kumari, **Dhruv Kumar**, Sundar Balasubramaniam, Navneet Goyal, Saiyedul Islam, and Jagat Sesh Challa. 2015. Parallelizing OPTICS for Commodity Clusters. In **ICDCN 2015**, 33:1–33:10.
- C38.** Poonam Goyal, Sonal Kumari, **Dhruv Kumar**, Sundar Balasubramaniam, and Navneet Goyal. 2014. Parallelizing OPTICS for Multicore Systems. In **ACM COMPUTE 2014**, 17:1–17:6.

Workshop Papers, Posters & Abstracts

- W1.** Vaibhav Devraj, **Dhruv Kumar**, Jagat Sesh Challa, Parth Agarwal, Navya Kommuri, Trizal Garg, Prisha Singhal, Dhruv Shah, and Yash Sinha. 2026. CricBench: A Multilingual Benchmark for Evaluating LLMs in Cricket Analytics. In **EMNLP 2026 Workshop (REALM)**.
- W2.** Parth Agarwal, Sanchit Bahuguna, and **Dhruv Kumar**. 2026. LLM Graders Are Systematically Harsher Than Industry Interviewers: A Benchmark of 221 Real Technical Interviews. In **EMNLP 2026 Workshop (REALM)**.
- W3.** Yash Sinha, Parth Shrikant Patil, Murari Mandal, and **Dhruv Kumar**. 2026. Beyond Accuracy: Diagnosing Algebraic Reasoning Failures in LLMs Across Nine Complexity Dimensions. In **EMNLP 2026 Workshop (MathNLP)**.
- W4.** Vedant Jawandhia, Yash Sinha, Ankan Pal, Murari Mandal, and **Dhruv Kumar**. 2026. Measuring Representation Robustness in Large Language Models for Geometry. In **ICML 2026 Workshop (AI4Math)**.
- W5.** Manan Gupta, Chinmay Pushkar, Sanchit Kabra, **Dhruv Kumar**, and Jagat Sesh Challa. 2026. MultiVulnBench: A Large-Scale Benchmark for Count Bias in LLM-Based Multi-Vulnerability Detection. In **ICML 2026 Workshop (CTB)**, also accepted at **AgenticUQ**, **CompLearn**, **FAGEN**, and **DL4C**.
- W6.** Ojaswi Prakash, **Dhruv Kumar**, Murari Mandal, Mohan Kankanhalli, and Yash Sinha. 2026. MafiaPersona: A Multi-Agent Adversarial Benchmark for Evaluating Persona Persistence in Large Language Models. In **ICML 2026 Workshop (NExT-Game)**.
- W7.** Prapti Patra and **Dhruv Kumar**. 2026. PoolBench: Benchmarking Large Language Models on Continuous Physical Action Selection in Eight-Ball Pool. In **ICML 2026 Workshop (NExT-Game)**.
- W8.** Preetham Sivalingam, Murari Mandal, **Dhruv Kumar**, and Saurabh Deshpande. 2026. TSQueryBench: LLM-as-a-Judge for Time-Series Explanations. In **ICML 2026 Workshop (FMSD)**.
- W9.** Moulik Gupta, Aaditya Jain, **Dhruv Kumar**, Murari Mandal, and Saurabh Deshpande. 2026. ReGeN: Reference-Guided Synthetic Multivariate Time Series Generation for Forecasting. In **ICML 2026 Workshop (FMSD)**.
- W10.** Atharva Gupta, **Dhruv Kumar**, Murari Mandal, and Saurabh Deshpande. 2026. Where Computation Lives Inside TabPFN: Causal Localisation of Attention Head Function. In **ICML 2026 Workshop (FMSD)**.
- W11.** Jay Agarwal, Siddharth Khare, and **Dhruv Kumar**. 2026. What Do Lorentz-Equivariant Jet Taggers Learn? In **ICML 2026 Workshop (AI4Physics)**.
- W12.** Shivansh Bibra, **Dhruv Kumar**, Murari Mandal, and Yash Sinha. 2026. Joint Evaluation of Compliance, Planning, and Consistency under Paraphrase: A Relational-Complexity View of Frontier LLMs. In **ICML 2026 Workshop**

(CTB).

- W13.** Jalluri Mahesh Kumar, Junjunoori Sri Chakri, Yash Kothari, Murari Mandal, Yash Sinha, and **Dhruv Kumar**. 2026. Evaluating LLM Reasoning on Operating System Algorithms via Step-Level Verification. In **ICML 2026 Workshop (CTB)**.
- W14.** Kanav Kapoor, Jagat Sesh Challa, Yash Sinha, Murari Mandal, and **Dhruv Kumar**. 2026. GraphStateEval: A Step-by-Step Evaluation Framework for Graph Algorithm Execution in Large Language Models via Intermediate State Tracing. In **ICML 2026 Workshop (CTB)**.
- W15.** Shubh Chapra, **Dhruv Kumar**, Murari Mandal, and Yash Sinha. 2026. The Complexity Ceiling Benchmark: A Multi-Domain Evaluation of Sequential Reasoning Under Depth Scaling. In **ICML 2026 Workshop (CTB)**.
- W16.** Aryan Khurana, Aravind Ramana RN, and **Dhruv Kumar**. 2026. Authority, Truth, and Citation Bias: A Large-Scale Multi-Domain Benchmark for Studying Epistemic Susceptibility in Large Language Models. In **ICML 2026 Workshop (EIML)**, also accepted at **AI4GOOD**.
- W17.** Agniva Maiti, Aditya Tiwari, Dwarikanath Mahapatra, Sudipta Roy, Yash Sinha, **Dhruv Kumar**, and Murari Mandal. 2026. Bridging the Visual Gap: Resource-Efficient VLM Adaptation for Meitei Mayek using Synthetic Multi-Modal Mixtures. In **ICML 2026 Workshop (GlobalSouthML)**.
- W18.** Agniva Maiti, Aarsh Muley, Aditya Tiwari, Dwarikanath Mahapatra, Sudipta Roy, Yash Sinha, **Dhruv Kumar**, and Murari Mandal. 2026. Evaluating Indic Literary Creativity In LLMs through the Lens of the Shastras. In **ICML 2026 Workshop (GlobalSouthML)**.
- W19.** Aviral Dawar, Roshan Karanth, Vikram Goyal, and **Dhruv Kumar**. 2026. IndicDB: Benchmarking Multilingual Text-to-SQL Capabilities in Indian Languages. In **ICML 2026 Workshop (Culture x AI)**, also accepted at **GlobalSouthML** and **ACL 2026 Workshop (MeLLM)**.
- W20.** Manya Pandey, **Dhruv Kumar**, Murari Mandal, and Saurabh Deshpande. 2026. GITCO: Gated Inference-Time Context Optimization in TSFMs. In **ICML 2026 Workshop (GlobalSouthML)**, also accepted at **FMSD**.
- W21.** Aaryan Nagpal, Debdeep Sanyal, Murari Mandal, **Dhruv Kumar**, and Saurabh Deshpande. 2026. Mix, Don't Pick: Why Synthetic Corpus Composition Matters for Time Series Foundation Model Pretraining. In **ICML 2026 Workshop (GlobalSouthML)**, also accepted at **FMSD** and **Forecast**.
- W22.** Karthik Sridhar, Atharva Gupta, Nishant Pradhan, Murari Mandal, **Dhruv Kumar**, and Saurabh Deshpande. 2026. Semantics or Structure? Auditing Text Sensitivity in Multimodal Time-Series Forecasting. In **ICML 2026 Workshop (GlobalSouthML)**, also accepted at **FMSD**.
- W23.** Manan Gupta and **Dhruv Kumar**. 2026. Diagnosing LLM Judge Reliability: Conformal Prediction Sets and Transitivity Violations. In **ICML 2026 Workshop (SCALE)**, also accepted at **CTB**, **SPIGM**, **FoGen**, and **FAGEN**.
- W24.** Manan Gupta and **Dhruv Kumar**. 2026. Is the Trust Region Necessary? Robust Entropy Adaptation Suffices for Continuous Control. In **ICML 2026 Workshop (SCALE)**.
- W25.** Manan Gupta and **Dhruv Kumar**. 2026. Latent Phase-Shift Rollback: Inference-Time Error Correction via Residual Stream Monitoring and KV-Cache Steering. In **ICML 2026 Workshop (SCALE)**, also accepted at **AgenticUQ**, **CTB**, **FoGen**, and **FAGEN**.
- W26.** Manan Gupta and **Dhruv Kumar**. 2026. Reasoning Phases Are Continuous, Not Discrete: Evidence from Switching Linear Dynamical Systems Applied to Chain-of-Thought Residual Streams. In **ICML 2026 Workshop (SCALE)**, also accepted at **CompLearn**, **FoGen**, and **Mech Interp**.
- W27.** Manan Gupta, Inderjeet Jayakumar Nair, Lu Wang, and **Dhruv Kumar**. 2026. Context Over Content: Exposing Evaluation Faking in Automated Judges. In **ICML 2026 Workshop (AgenticUQ)**, also accepted at **SPIGM**, **FoGen**, **FAGEN**, and **AI4GOOD**.
- W28.** Manan Gupta and **Dhruv Kumar**. 2026. DPMI: A Principled Index for Neural Polysemanticity via Dirichlet Process Mixture Modeling. In **ICML 2026 Workshop (SCALE)**, also accepted at **CompLearn** and **FoGen**.
- W29.** Aaryaman Kalani, **Dhruv Kumar**, Mohan Kankanhalli, Murari Mandal, and Yash Sinha. 2026. RASE: Retain-Agnostic Machine Unlearning via Activation Subspace Projection. In **CVPR 2026 Workshop (MUV)**.
- W30.** Shubham Mishra, Shiv Tiwari, Samyek Jain, Gorang Mehrishi, **Dhruv Kumar**, Pratik Narang, and Harsh Sharma. 2026. From Facts to Conclusions: Integrating Deductive Reasoning in Retrieval-Augmented LLMs. In **ICLR 2026 Workshop on LLM Reasoning. arXiv (2025)**.
- W31.** Debdeep Sanyal, Aaryan Nagpal, **Dhruv Kumar**, Murari Mandal, and Saurabh Deshpande. 2025. time2time: Causal Intervention in Hidden States to Simulate Rare Events in Time Series Foundation Models. In **NeurIPS 2025 Workshop on Time Series Foundation Models (BERT4S)**.
- W32.** Sankalp Tattwadharshi Swain, Anshika Krishnatray, **Dhruv Kumar**, and Jagat Sesh Challa. 2025. Talking with Oompa Loompas: A Novel Framework for Evaluating Linguistic Acquisition of LLM Agents. In **NeurIPS 2025 Workshop (WiML)**, also accepted at **LLM Evaluation**.
- W33.** Utkarsh Tiwari, Aryan Seth, Adi Mukherjee, Kaavya Mer, Kavish, and **Dhruv Kumar**. 2025. DebateBench: A Challenging Long Context Reasoning Benchmark For Large Language Models. In **NAACL SRW 2025**.
- W34.** Shreya Bhatia, Tarushi Gandhi, **Dhruv Kumar**, and Pankaj Jalote. 2024. Unit Test Generation using Generative AI: A Comparative Performance Analysis of Autogeneration Tools. In **LLM4Code@ICSE 2024**, 54–61.
- W35.** **Dhruv Kumar**, Sohaib Ahmad, Abhishek Chandra, and Ramesh K. Sitaraman. 2023. AggFirstJoin (Workshop Version). In **CCGridW 2023**, 308–310.

- W36.** Dhruv Kumar, Joel Wolfrath, Abhishek Chandra, and Ramesh K. Sitaraman. 2022. Towards WAN-Aware Join Sampling over Geo-Distributed Data. In **EdgeSys@EuroSys 2022**, 13–18.
- W37.** Yuanli Wang, Joel Wolfrath, Nikhil Sreekumar, Dhruv Kumar, and Abhishek Chandra. 2021. Accelerated Training via Device Similarity in Federated Learning. In **EdgeSys@EuroSys 2021**, 31–36.
- W38.** Yuanli Wang, Dhruv Kumar, and Abhishek Chandra. 2020. Exploiting Data Heterogeneity for Performance and Reliability in Federated Learning (Poster). In **SEC 2020**, 164–166.
- W39.** Dhruv Kumar, Aravind Alagiri Ramkumar, Rohit Sindhu, and Abhishek Chandra. 2019. DeCaf: Iterative Collaborative Processing over the Edge. In **HotEdge 2019**.
- W40.** Dhruv Kumar, Jian Li, Abhishek Chandra, and Ramesh K. Sitaraman. 2019. A TTL-Based Approach for Data Aggregation in Geo-Distributed Streaming Analytics (Abstract). In **SIGMETRICS 2019**, 15–16.

Preprints (arXiv / SSRN)

- P1.** Rakshita Goel, S. Pranav Kumar, Anmol Agrawal, Divyan Poddar, Pratik Narang, and Dhruv Kumar. 2026. Domain-Partitioned Hybrid RAG for Legal Reasoning: Toward Modular and Explainable Legal AI for India. **arXiv (2026)**.
- P2.** Gaurish Trivedi, Alakh Sharma, Kartikey Singh Bhandari, Yash Sinha, Pratik Narang, Dhruv Kumar, and Jagat Sesh Challa. 2026. Trust Regions Sell, But Who’s Buying? Overlap Geometry as an Alternative Trust Region for Policy Optimization. **arXiv (2026)**.
- P3.** Devanshu Sahoo, Manish Prasad, Vasudev Majhi, Arjun Neekhara, Yash Sinha, Murari Mandal, Vinay Chamola, and Dhruv Kumar. 2026. The Compliance Paradox: Semantic-Instruction Decoupling in Automated Academic Code Evaluation. **arXiv (2026)**.
- P4.** Aditya Thole, Anmol Agrawal, Arnav Ramamoorthy, and Dhruv Kumar. 2026. PhysicsSolutionAgent: Towards Multimodal Explanations for Numerical Physics Problem Solving. **arXiv (2026)**.
- P5.** Kartikey Singh Bhandari, Manav Ganesh, Yashwant Viswanathan, Archit Agrawal, Dhruv Kumar, and Pratik Narang. 2026. Actionable Advice from Reviews via Mixture of LoRA Experts: A Two-LLM Pipeline for Issue Extraction and Business Recommendations. **arXiv (2026)**.
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Framework for Automated Code Review. **SSRN (2025)**.

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- P30.** Atharva Mehta, Nipun Gupta, Aarav Balachandran, **Dhruv Kumar**, Pankaj Jalote, et al. 2023. Can ChatGPT Play the Role of a Teaching Assistant in an Introductory Programming Course? **arXiv (2023)**.

Sponsored Research Projects

1. SPANDAN — AI/ML for Strengthening Police Investigation Processes	
Funding Agency	Gujarat Police (Office of Superintendent of Police, Ahmedabad Rural)
Budget	INR 34,80,000
Duration	11 months (03/2026–01/2027)
Status	Ongoing
Role	PI

2. Revolutionizing Indian Higher Education Institutes with Generative AI	
Funding Agency	BITS Pilani
Budget	INR 20,00,000
Duration	24 months (11/2024–10/2026)
Status	Ongoing
Role	PI

Summary of Results & Publications:

- Pathak, A.; **Kumar, D.**; et al. 2025. Rubric Is All You Need: LLM-based code evaluation. *ACM ICER 2025*.
- Arora, U.; **Kumar, D.**; et al. 2025. Analyzing LLM usage in an advanced computing class in India. *ACE '25*, 154–163.

3. AI for River–Climate Modeling	
Funding Agency	The Nature Conservancy (NGO)
Budget	INR 25,00,000
Duration	11 months (08/2024–06/2025)
Status	Completed
Role	Co-PI

Summary of Results & Publications:

- Rallapalli, S.; **Kumar, D.**; et al. 2025. Quantifying the role of land use–land cover for adaptive river basin management. *Journal of Environmental Management* (under submission).
- Rallapalli, S.; **Kumar, D.**; et al. 2025. Targeting degraded hotspots in riparian corridors for rehabilitation. *Journal of Environmental Management* (under submission).
- Rallapalli, S.; **Kumar, D.**; et al. 2025. Quantifying the synergistic influence of LULC and climate on hydro-climatological dynamics. *Journal of Hydrology* (under submission).
- Rallapalli, S.; **Kumar, D.**; et al. 2025. Disaggregating the influence of climate change and LULC on streamflow. *Journal of Environmental Management* (under submission).

4. Generative AI in Education Grant	
Funding Agency	WISE–IIE Higher Education Consortium
Budget	INR 30,00,000
Duration	10 months (08/2024–05/2025)
Status	Completed
Role	Co-PI

Summary of Results & Publications:

- Akolekar, H.; **Kumar, D.**; et al. 2025. Generative AI tools in mechanical engineering education. *Scientific Reports* 15, 9214.
- Gupta, R.; **Kumar, D.**; et al. 2025. Saksham AI: Socratic tutoring for coding education in India. *ACM JCSS 2026*.
- Goyal, H.; **Kumar, D.**; et al. 2025. Impact of LLMs on K-12 education in rural India. *Scientific Reports 2025*.
- Sanyal, D.; **Kumar, D.**; et al. 2025. Investigating pedagogical teacher and student LLM agents. *EMNLP 2025*.

5. Google Cloud Education Grant	
Funding Agency	Google Cloud
Budget	INR 12,50,000
Duration	12 months (01/2024–12/2024)
Status	Completed
Role	PI